Applications of Hydroacoustics

From marine ecology to geophysical hazards

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Learning objectives

By the end of this unit, you should be able to:

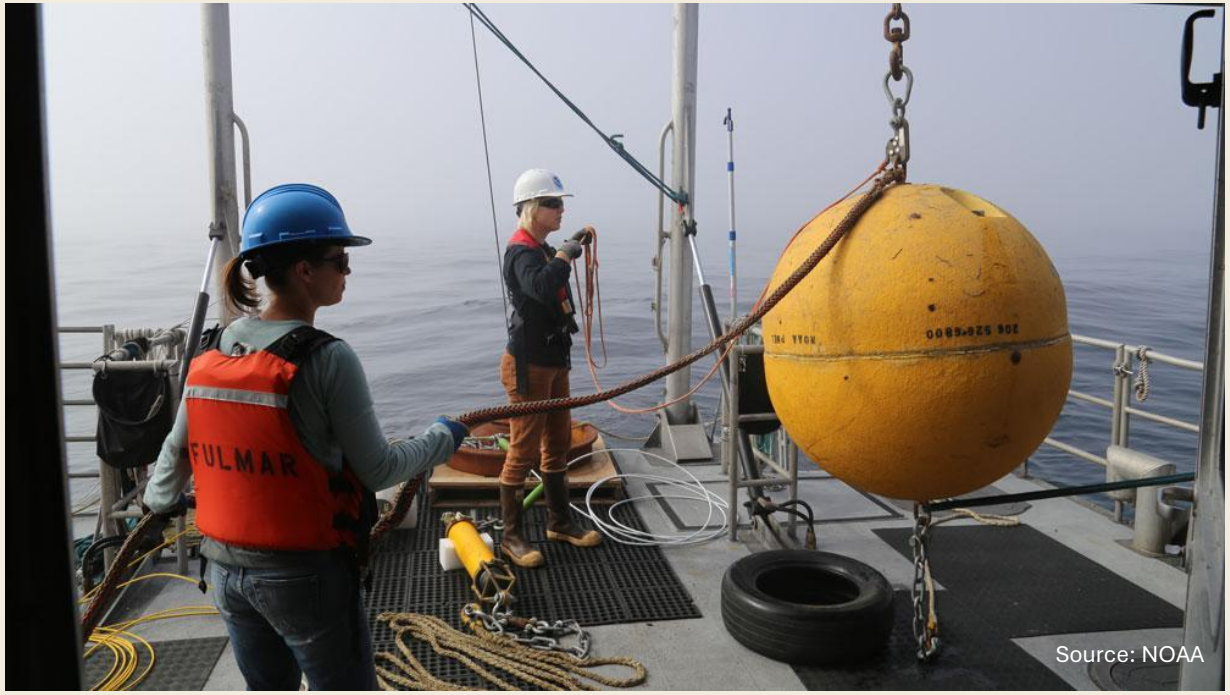
- Describe major applications of hydroacoustic monitoring.
- Interpret acoustic evidence in an applied context.
- Explain strengths and limitations of passive acoustic monitoring.
- Connect signal characteristics to management or hazard-monitoring questions.

Why monitor the ocean acoustically?

Hydroacoustic monitoring can provide:

- large spatial coverage
- continuous observations
- monitoring in remote environments
- detection without direct visual observation
- observations at night and in poor visibility

A hydrophone can continuously monitor sounds produced by animals, human activities and geophysical processes, including at night or when direct visual observations are impossible.



Source: NOAA


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Application 1: Passive acoustic monitoring of marine mammals

Hydrophones can be used to:

- Detect when vocalising animals are present.
- Identify characteristic calls of some species.
- Investigate daily and seasonal calling patterns.
- Study acoustic behaviour.
- Estimate movement or location when suitable acoustic arrays and methods are available.

Important

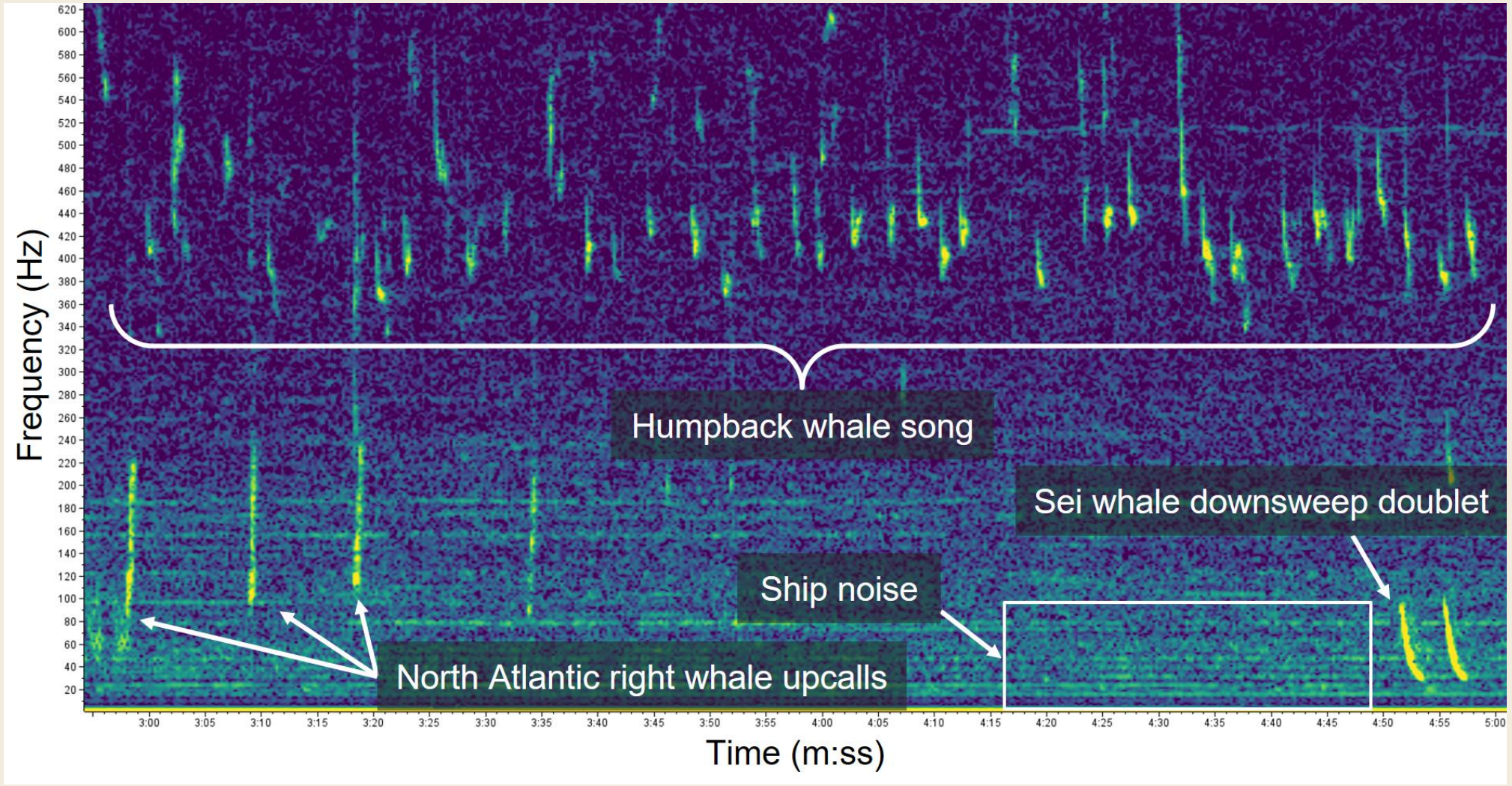
A hydrophone detects vocalising animals, not every animal present.

NOAA has used non-invasive passive acoustic monitoring for marine mammals since 2006.



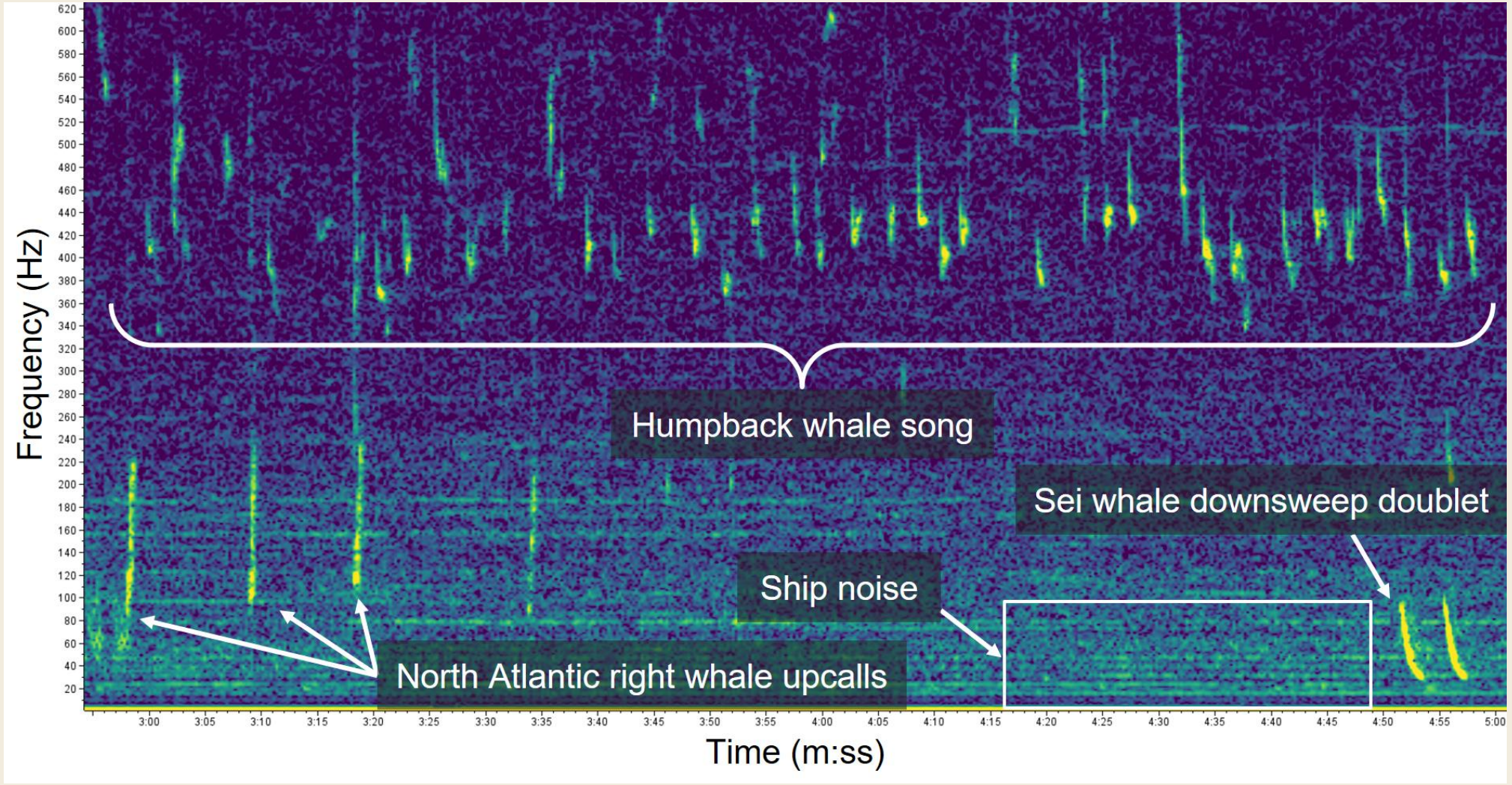
Two North Atlantic right whales interacting with a pod of pilot whales (MMPA Research Permit # 775-1875). Credit: NOAA Fisheries https://www.fisheries.noaa.gov/new-england-mid-atlantic/science-data/marine-mammal-acoustics

Spectrogram showing unique calls by multiple species including humpback whale song, North Atlantic right whale upcalls, and a sei whale downsweep doublet with low-frequency ship noise overlapping. Credit: NOAA Fisheries/ https://www.fisheries.noaa.gov/new-england-mid-atlantic/science-data/marine-mammal-acoustics



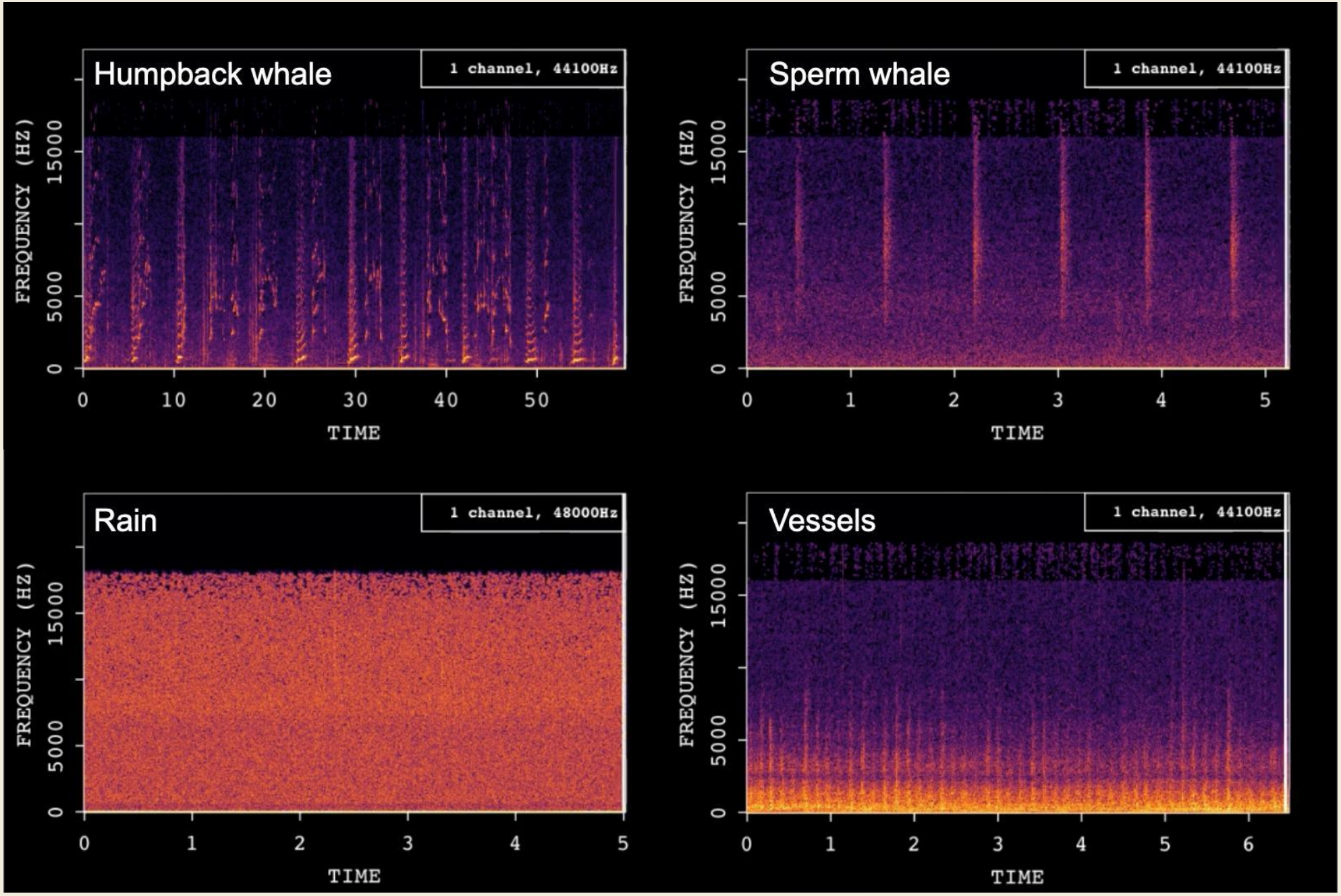
Look at the spectrogram: Can you distinguish the right whale upcalls, humpback whale song and ship noise? Which acoustic signals might overlap in frequency?

Spectrogram showing unique calls by multiple species including humpback whale song, North Atlantic right whale upcalls, and a sei whale downsweep doublet with low-frequency ship noise overlapping. Credit: NOAA Fisheries/ https://www.fisheries.noaa.gov/new-england-mid-atlantic/science-data/marine-mammal-acoustics



Interpretation

- North Atlantic right whale upcalls appear as short, rising-frequency signals.
- Humpback whale song consists of multiple structured vocalisations.
- Sei whale downsweeps appear as descending-frequency traces.
- Ship noise produces sustained energy at low frequencies.



Spectrograms illustrating the acoustic characteristics of humpback whale vocalisations, sperm whale clicks, rainfall and vessel noise, recorded in Hawaiian Islands Humpback Whale National Marine Sanctuary. Source: NOAA–U.S. Navy SanctSound Project. https://sanctsound.ioos.us/q_what-hear.html

Self-check:

Question: Why might ship noise make whale calls harder to detect?

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Question: Why might ship noise make whale calls harder to detect?

Answer: When ship noise overlaps with whale calls in frequency and time, it can mask them, reducing their detectability.

Case study — What can repeated detections tell us?

Imagine a hydrophone detects the same whale call repeatedly over several months.

We could investigate:

- when whales are acoustically present
- seasonal patterns in occurrence
- changes in calling activity
- timing of migration or habitat use

But acoustics alone may not tell us:

- the exact number of whales present
- whether silent whales are nearby
- age or sex unless the call type is diagnostic
- why the animals are present

Acoustic detection ≠ number of animals

A single whale can produce many calls, while several whales may remain silent. Therefore, the number of recorded calls cannot normally be converted directly into the number of whales present.

Strengths and limitations of passive acoustic monitoring

Strengths	Limitations
Non-invasive	Silent animals are not detected
Works at night	Detection range varies
Works in poor visibility	Species identification may be uncertain
Can record continuously	Presence does not equal abundance
Suitable for long deployments	Background noise can mask calls
Useful in remote areas	Equipment and data processing are required

From Whale Detection to Management

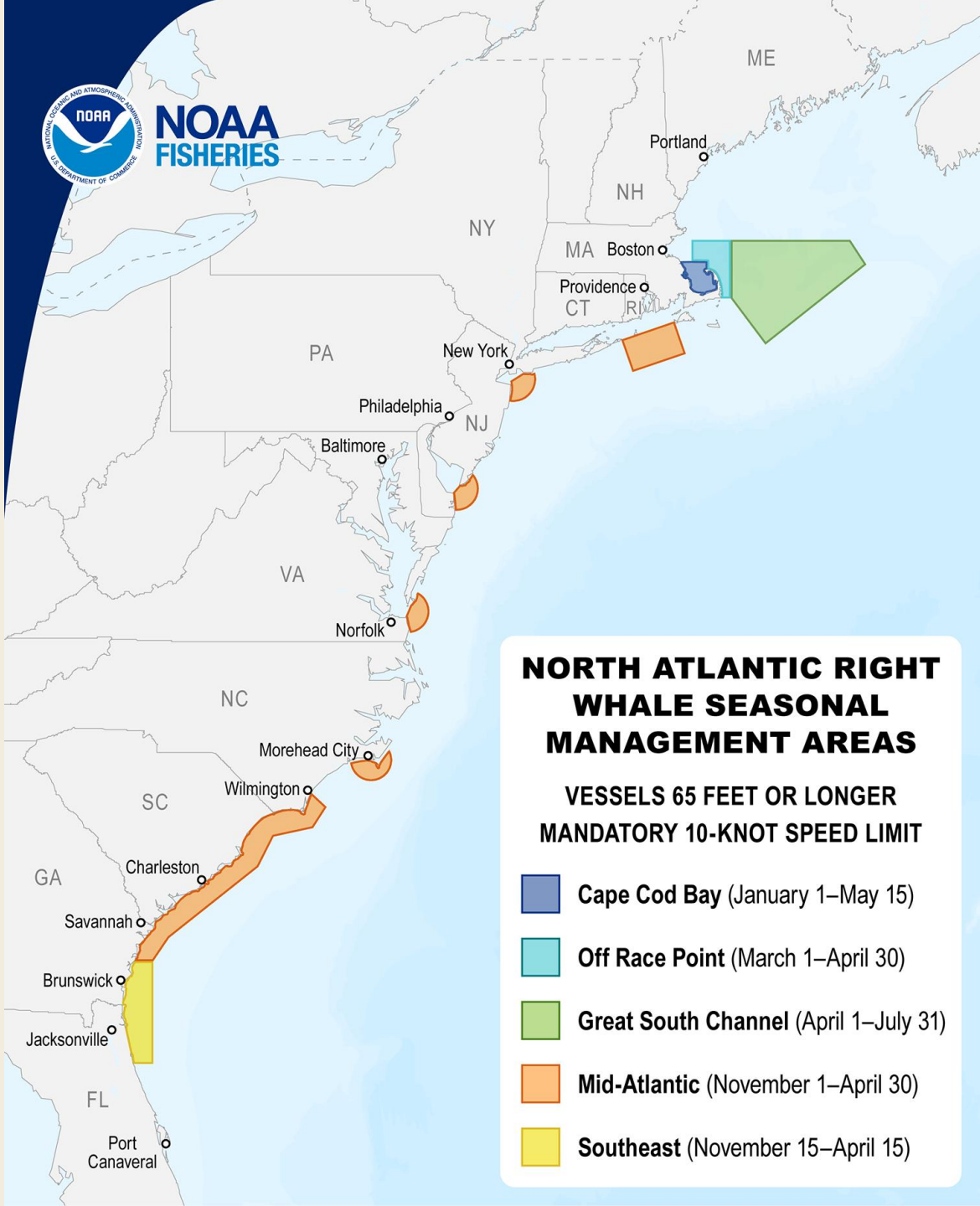
Passive acoustic monitoring can directly inform conservation decisions.

Protecting endangered North Atlantic right whales

NOAA uses acoustic detections to help establish temporary Right Whale Slow Zones, where vessel operators are encouraged to reduce speed to 10 knots or less. NOAA also establishes Seasonal Management Areas, where most vessels 65 feet or longer must follow mandatory 10-knot speed restrictions during designated periods.

Why?

Reducing vessel speed helps lower the risk of serious or fatal whale–ship collisions. Hydroacoustic observations can support real-world conservation and management decisions.



North Atlantic right whale Seasonal Management Areas along the US East Coast, showing regions with mandatory seasonal vessel-speed restrictions. Source: NOAA Fisheries.

Human-generated underwater noise

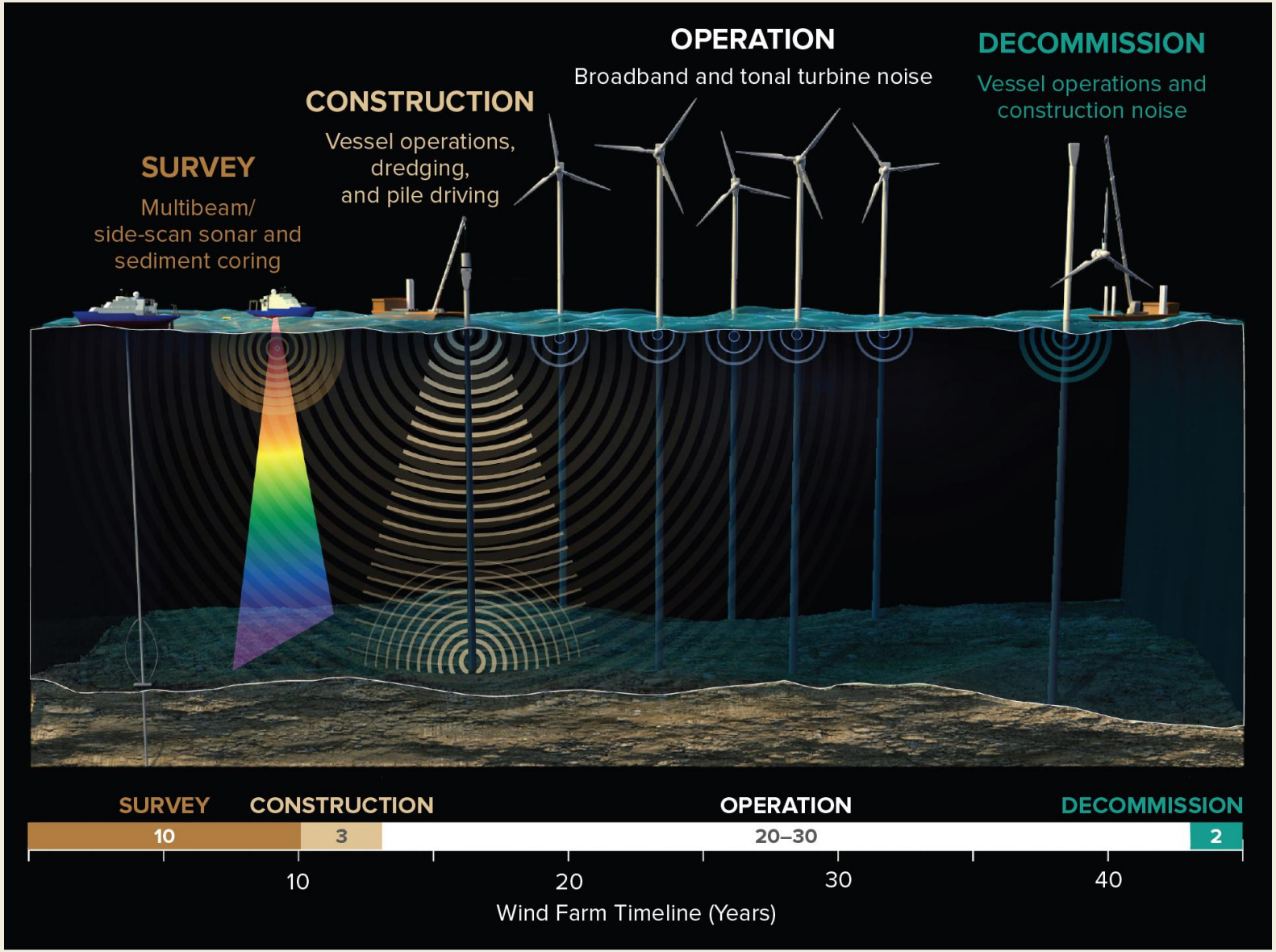
Common sources include:

- Shipping and recreational vessels.
- Sonar systems.
- Seismic surveys.
- Pile driving and offshore construction.

Potential effects on marine life

- Acoustic masking: Noise interferes with the detection of biologically important sounds.
- Behavioural disturbance: Animals may change their calling, movement or feeding behaviour.
- Habitat displacement: Repeated disturbance may alter habitat use.
- Physiological effects: Intense exposure can cause temporary or permanent hearing damage in susceptible animals.

Effects depend on sound characteristics, exposure duration and species sensitivity.



Underwater sound sources during the different phases of an offshore wind farm's lifetime. Source: Mooney et al. (2020), Figure 1, CC BY 4.0.

Case study — When ship noise overlaps with whale communication

A passing vessel can substantially increase underwater sound levels across particular frequency bands. If vessel noise overlaps with whale calls, it can reduce the distance over which animals can communicate.

This is called acoustic masking.

Link:
https://pmel.noaa.gov/acoustics/env-noise/shipover.html

Activity: Listen to a Passing Boat

Open the NOAA small boat recording:

1. Listen to how the vessel noise changes as the boat approaches and passes the hydrophone.
2. Identify when the recorded sound is strongest.
3. Examine the spectrogram. Which frequency ranges contain the strongest energy?
4. Consider how this noise might mask marine mammal calls.

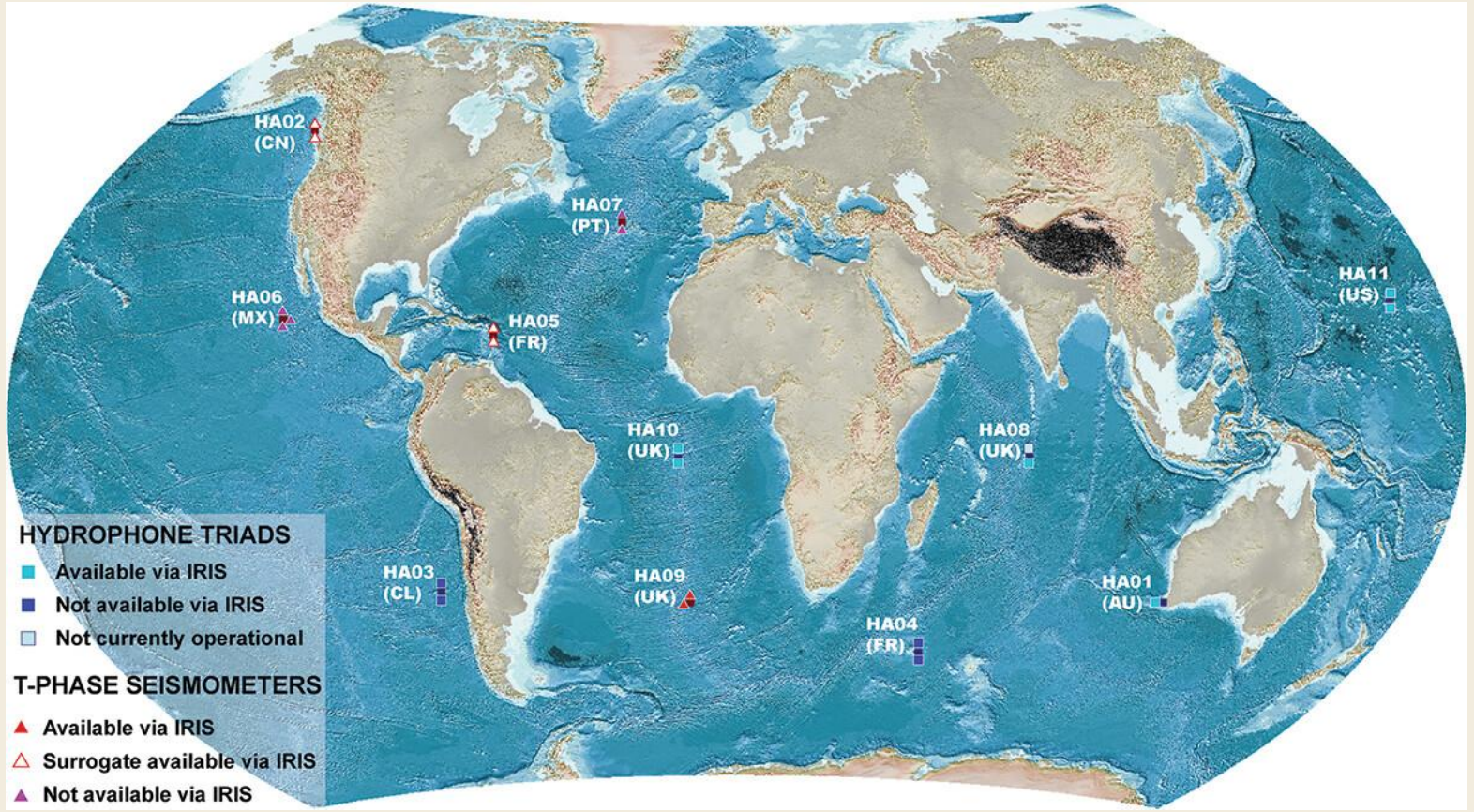
Note: The audio has been played at ten times its original speed, changing its perceived pitch and duration. During the closest vessel passage, the recording becomes saturated. Saturation occurs when the signal exceeds the recording system's measurement range, causing distortion. The recording is therefore useful for qualitative interpretation, but its saturated section cannot be used for reliable quantitative measurement of the original sound amplitude.

Hydroacoustics for geophysical monitoring

Hydrophones can detect acoustic signals generated by:

- submarine earthquakes
- volcanic eruptions
- landslides
- Explosions

Low-frequency sound can travel very long distances through the ocean, allowing events to be detected far from their source.



Source: CTBTO — Hydroacoustic Monitoring

Did you know?

The CTBTO operates a global hydroacoustic monitoring network consisting of:

- **6 hydrophone stations:** Underwater sensors that detect acoustic pressure fluctuations.
- **5 T-phase stations:** Seismic sensors on islands that detect underwater acoustic energy converted into seismic waves.

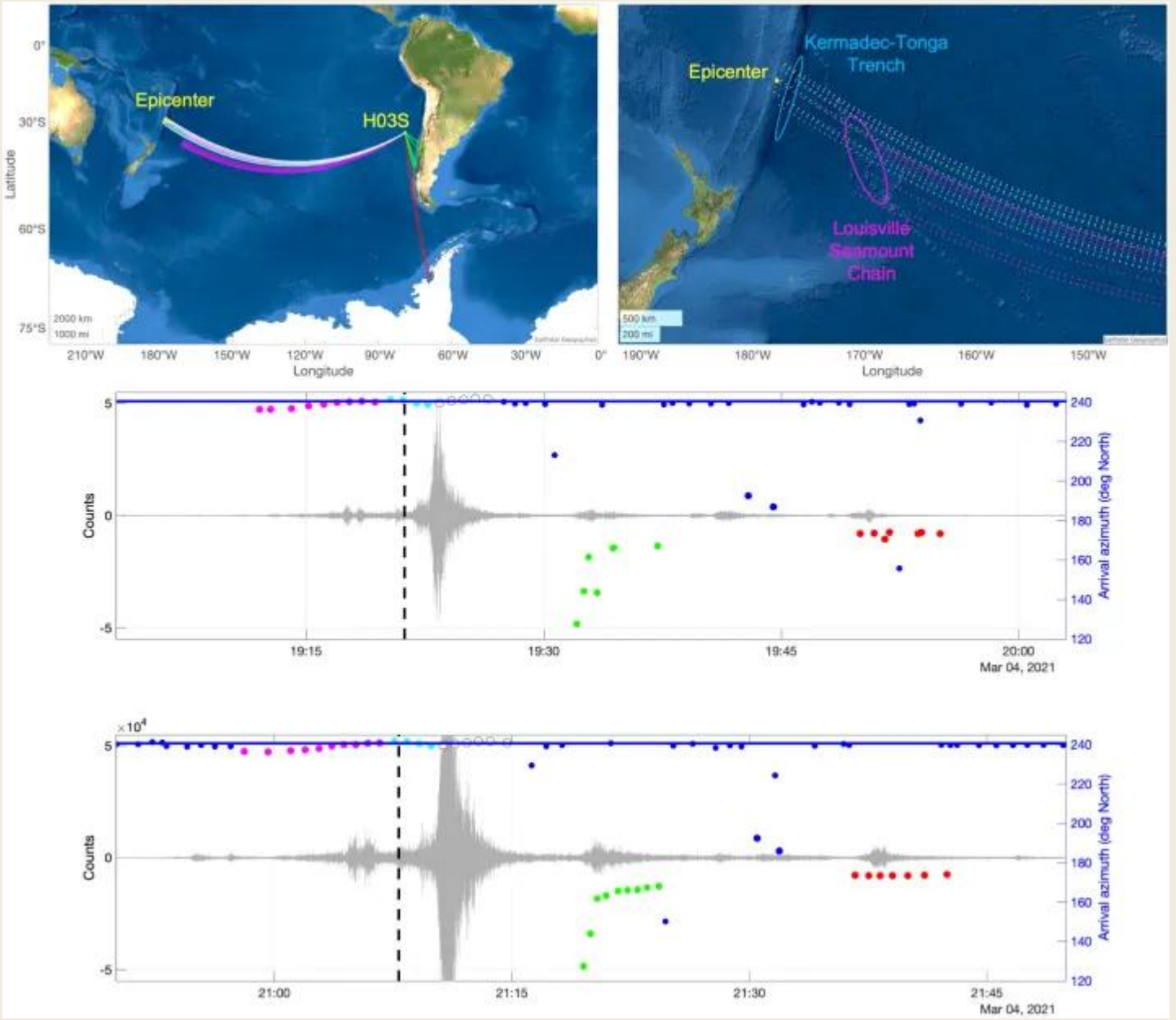
Under favourable conditions, low-frequency underwater sound can propagate thousands of kilometres, allowing distant stations to detect natural events and potential underwater nuclear explosions.

How can a hydrophone detect an earthquake?

1. An earthquake generates seismic waves and seafloor motion.
2. Seafloor motion transfers part of its energy into the water as acoustic waves.
3. Sound propagates through the ocean, sometimes over very long distances.
4. A distant hydrophone records the arriving acoustic signal.

These signals are often called T-waves or T-phases and can complement land-based seismic observations.

Did you know? Earthquake-generated acoustic signals have been detected more than 15,000 km from their source, demonstrating the extraordinary range of underwater sound propagation.



Hydroacoustic T-wave signals generated by submarine earthquakes and recorded at distant CTBTO hydrophones. Source: Oliveira et al. (2022), Figure 3, CC BY 4.0.

One network, multiple scientific uses

Application	Typical source	Example scientific question
Marine mammals	Biological	Which species are present and when?
Shipping noise	Anthropogenic	How often and how strongly are animals exposed to vessel noise?
Earthquakes	Geophysical	When and where did the event occur?
Volcanic activity	Geophysical	Is a submarine volcano acoustically active?
Hazard monitoring	Geophysical	Can acoustic observations improve understanding of a source event?
Nuclear-test monitoring	Anthropogenic	Did an underwater explosion occur?

Notebook 3 — Applied Hydroacoustics

You will analyse a real hydroacoustic recording in an applied context.

Workflow

- Load the recording.
- Inspect the waveform.
- Calculate the frequency spectrum.
- Generate a spectrogram.
- Identify the likely source.
- Quantify simple signal characteristics.
- Interpret the result in an ecological or geophysical context.
- Discuss limitations and uncertainty.

Your task: Use acoustic evidence to investigate a real marine or geophysical event. Describe what the recording reveals, explain how the results could inform monitoring or management, and identify what cannot be determined from the acoustic data alone.

Take-home messages

- The same hydrophone can address very different scientific questions.
- Passive acoustics can monitor animals, human noise and geophysical events.
- Long-range propagation makes hydroacoustics particularly useful in remote oceans.
- Acoustic observations are most powerful when combined with other data.
- Interpretation depends on context, propagation and detection conditions.
- Uncertainty matters in both ecological and hazard applications.

Explore further

Marine mammals

https://www.fisheries.noaa.gov/new-england-mid-atlantic/science-data/marine-mammal-acoustics

Anthropogenic sound

https://www.fisheries.noaa.gov/national/science-data/sounds-ocean-environmental-and-anthropogenic

Earthquakes / global monitoring

https://www.ctbto.org/our-work/monitoring-technologies/hydroacoustic-monitoring

Tsunami warning

https://www.noaa.gov/explainers/us-tsunami-warning-system

Reference

Scientific publications

- Mooney, T. A., Andersson, M. H., & Stanley, J. (2020). Acoustic impacts of offshore wind energy on fishery resources. *Oceanography, 33*(4), 82–95. https://doi.org/10.5670/oceanog.2020.408
- Oliveira, T. C. A., et al. (2022). Megameter propagation and correlation of T-waves from Kermadec Trench and Islands. *Frontiers in Marine Science, 9*, 1009013. https://doi.org/10.3389/fmars.2022.1009013

Scientific and educational resources

- NOAA Fisheries. *Marine Mammal Acoustics*. https://www.fisheries.noaa.gov/new-england-mid-atlantic/science-data/marine-mammal-acoustics
- NOAA–U.S. Navy. *SanctSound: What Did We Hear?* https://sanctsound.ioos.us/q_what-hear.html
- NOAA Fisheries. *Reducing Vessel Strikes to North Atlantic Right Whales*. https://www.fisheries.noaa.gov/national/endangered-species-conservation/reducing-vessel-strikes-north-atlantic-right-whales
- NOAA PMEL. *Small Boat Passing a Hydrophone*. https://pmel.noaa.gov/acoustics/env-noise/shipover.html
- CTBTO. *Hydroacoustic Monitoring*. https://www.ctbto.org/what-we-do/monitoring-technologies/hydroacoustic-monitoring

Figure and media credits

- Slide 3: Hydrophone deployment photographs. NOAA.
- Slides 4–5: Marine mammal photograph and multispecies spectrogram. NOAA Fisheries.
- Slide 6: Comparative underwater sound spectrograms. NOAA–U.S. Navy SanctSound Project.
- Slide 11: North Atlantic right whale Seasonal Management Areas. NOAA Fisheries.
- Slide 12: Offshore wind farm underwater sound sources. Mooney et al. (2020), Figure 1, CC BY 4.0.
- Slide 14: International Monitoring System hydroacoustic station map. CTBTO.
- Slide 15: Earthquake-generated T-wave propagation and recordings. Oliveira et al. (2022), Figure 3, CC BY 4.0.